

**Note:** The following exercises assume that an `int` is 4 bytes.

**Resuelva los siguientes ejercicios a mano y descubra qué imprime cada fragmento de código en la consola. Puedes verificar tus respuestas ejecutando tu código en tu PC**

**1. Exercise (Little Endian):**

C/C++

```
int main() {  
    int num = 0x41424344; // ASCII for 'ABCD'  
    unsigned char *p = (unsigned char *)&num;  
    printf("%c\n", *p);  
    return 0;  
}
```

**2. Exercise (Big Endian):**

C/C++

```
int main() {  
    int num = 0x41424344; // ASCII for 'ABCD'  
    unsigned char *p = (unsigned char *)&num;  
    printf("%c\n", *p);  
    return 0;  
}
```

**3. Exercise (Little Endian):**

C/C++

```
int main() {  
    int num = 0x45464748; // ASCII for 'EFGH'  
    unsigned char *p = (unsigned char *)&num + 2;  
    printf("%c\n", *p);  
    return 0;  
}
```

#### 4. Exercise (Big Endian):

C/C++

```
int main() {  
    int num = 0x45464748; // ASCII for 'EFGH'  
    unsigned char *p = (unsigned char *)&num + 1;  
    printf("%c\n", *p);  
    return 0;  
}
```

#### 5. Exercise (Little Endian):

C/C++

```
int main() {  
    int num = 0x494A4B4C; // ASCII for 'IJKL'  
    unsigned char *p = (unsigned char *)&num + 3;  
    printf("%c\n", *p);  
    return 0;  
}
```

#### 6. Exercise (Big Endian):

C/C++

```
int main() {  
    int num = 0x494A4B4C; // ASCII for 'IJKL'  
    unsigned char *p = (unsigned char *)&num + 2;  
    printf("%c\n", *p);  
    return 0;  
}
```

#### 7. Exercise (Big Endian):

C/C++

```
int main() {  
    int num = 0x12345678;  
    unsigned char *p = (unsigned char *)&num;  
    printf("%x\n", *p);  
    return 0;  
}
```

## 8. Exercise

```
C/C++
int arr[5] = {10, 20, 30, 40, 50};
int *p = arr;
printf("%d\n", *(p + 4 - 2));
```

## 9. Exercise(Little Endian):

```
C/C++
#include <stdio.h>
typedef struct {
    int first_number;
    int second_number;
} MyStruct;

int main() {
    MyStruct my_struct;
    char *ptr = (char*) &my_struct;

    my_struct.first_number = 0x11223344;
    my_struct.second_number = 0xAABBCCDD;

    printf("%02X", *(ptr + 7));
}
```

Les comparto el [video](#) de como solucionar el número 9.

## 10. Exercise(Little Endian):

```
C/C++
#include <stdio.h>

typedef struct {
    char a;
    int b;
    short c;
} MyNewStruct;
```

```

int main() {
    MyNewStruct s;
    char *ptr = (char*) &s;

    s.a = 0x7F;
    s.b = 0x11223344;
    s.c = 0x8001;

    printf("%02X", (int)*(short*)(ptr + 5));
}

```

## 11. Exercise

```

C/C++
int val = 25;
int *ptr1 = &val;
int *ptr2 = &val;
int *ptr3 = ptr1;
*ptr3 = 30;
*ptr2 = 40;
printf("%d\n", val);

```

## 12. Exercise

```

C/C++
int dato = 50;
int *a = &dato;
int *b = &dato;
*a -= 10;
*b -= 5;
printf("%d\n", dato);

```